

# Safe Few-Shot World-Model Adaptation to Unseen Joint-Locking Failures

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**Abstract**—Robot world models deployed after hardware damage must adapt to unseen interaction dynamics while exactly respecting the altered mechanism. We study planar pushing with a serial manipulator whose diagnosed joint is locked at a known angle, while friction, delay, payload, and contact effects remain unknown. We introduce a contact-aware physical-context rollout-risk intervention-projected world model (PC-RR-IPWM). It combines analytic lock projection, a frozen contact-aware robot transition, a shared object model, explicit pusher geometry, and a support-aware low-rank intervention residual. An observable physical-context posterior inferred from 25 safe transitions modulates the residual through an exact-zero centered gate and a delayed rollout-risk schedule. Against a compute-, data-, and adapter-matched shared model, object rollout RMSE improves in all 18 combinations of three seeds, two held-out domains, and H10/H25/H50, from 2.04% to 37.84%. Lock violation is zero and controls remain practically equivalent. **[REAL-ROBOT TODO: Add real-arm result.]**

## I. INTRODUCTION

A locked joint changes the robot state manifold discretely, while contact dynamics vary continuously with friction, backlash, delay, and payload. An unconstrained model can violate the known lock; fitting a complete model after failure is data-intensive and unsafe. Fault-tolerant methods adapt behavior [1], [2], and rapid adaptation infers latent properties from recent transitions [3], [4]. Pushing models learn uncertain contact dynamics [5], [6], [7], [8], but typically assume an intact robot or do not enforce the damaged manifold.

We propose PC-RR-IPWM, which assigns known damage to analytic projection and allows unknown physics to modify only an unsupported-intervention object residual. Projected pusher geometry enters the object block through a stop-gradient bridge. A centered context gate gives exact base recovery, and a depth schedule limits accumulated residual error.

Our contributions are: (i) exact projection plus contact-aware block-coordinate dynamics; (ii) a support-aware geometry/latent object intervention; (iii) an observable physical-context gate with reversible fallback; and (iv) a strict three-seed matched evaluation and component ablations.

## II. RELATED WORK

**Fault adaptation.** Damage recovery has been approached by repertoire search [1], adversarial fault training [2], latent extrinsics [3], residual simulator tuning [4], and meta-learned adaptive control [9]. We instead adapt a predictive manipulation model after a diagnosed discrete topology change.

**Contact dynamics.** Learned forward models support visual and model-based planning [5], [10]. Few-sample GP models [6], hybrid online identification [11], recurrent sim-to-real adaptation [8], and SE(2)-equivariant pushing models [7] address object physics. Our structural prior instead couples exact damaged-robot feasibility to explicit pusher geometry.

## III. PROBLEM FORMULATION

State  $s_t = [q_t, \dot{q}_t, o_t]$  contains five joint positions, velocities, and planar object state;  $a_t \in \mathbb{R}^5$ . Diagnosis gives lock  $d = (m, \bar{q})$ . Unknown physical context  $z \in \mathbb{R}^8$  is inferred from  $K$  safe transitions. D3 is held out from D1/D2/D4/D5 training support and is combined with held-out physics. We minimize autonomous H10/H25/H50 error while requiring  $q_t^j = \bar{q}^j, \dot{q}_t^j = 0$  for locked coordinates.

## IV. METHOD

### A. Analytic projection

$$\Pi_d(q)_j = (1 - m_j)q_j + m_j\bar{q}_j, \quad (1)$$

$$\Pi_d(\dot{q})_j = (1 - m_j)\dot{q}_j, \quad \Pi_d(a)_j = (1 - m_j)a_j. \quad (2)$$

Projection is applied before and after prediction, giving exact feasibility.

### B. Contact-aware block-coordinate model

A recurrent robot block predicts free-joint transitions from projected state, action, and contact/object context. Contact input is retained because deleting it destabilizes free-joint pushing rollouts. Forward kinematics produces pusher geometry. A shared object base receives detached robot features, preventing object loss from rewriting the frozen robot block.

### C. Support-aware intervention

For training lock support  $\mathcal{S}$ ,

$$\nu(d) = \mathbb{I}[m \notin \mathcal{S}] \mathbb{I}[\|m\|_1 > 0].$$

The intact term forces bypass. Object prediction is

$$\hat{o}_{t+1} = f_o(o_t, \text{sg}(h_{t+1}^r)) + \nu(d)\alpha_t(z)(r_g + r_\ell),$$

where  $r_g$  uses explicit pusher displacement/contact geometry and  $r_\ell$  is a low-rank latent residual.

TABLE I  
OBJECT RMSE IMPROVEMENT (%) OVER THE MATCHED BASELINE.

| Seed | Domain       | H10   | H25   | H50   |
|------|--------------|-------|-------|-------|
| 7    | composition  | 12.27 | 5.33  | 37.84 |
| 7    | mixed-unseen | 15.89 | 32.50 | 36.60 |
| 17   | composition  | 6.53  | 13.54 | 12.52 |
| 17   | mixed-unseen | 9.10  | 15.50 | 2.95  |
| 27   | composition  | 7.62  | 2.04  | 18.98 |
| 27   | mixed-unseen | 9.19  | 2.85  | 2.58  |

TABLE II  
SEED17 D3 MIXED-UNSEEN OBJECT IMPROVEMENT (%).

| Variant         | H10  | H25   | H50   |
|-----------------|------|-------|-------|
| Full            | 9.10 | 15.50 | 2.95  |
| No geometry     | 4.57 | 12.36 | 1.17  |
| No latent       | 3.54 | 2.50  | 3.32  |
| No intervention | 0.01 | 0.09  | 0.41  |
| No depth risk   | 3.14 | 13.98 | -3.86 |

#### D. Context and rollout risk

An amortized posterior estimates 8D context from state/action transitions without simulator labels. The frozen rule uses K25 and scale 1.38. Centering  $g(z) = h(z) - h(0)$  guarantees exact K0 recovery. Residual modulation has an eight-step grace period and  $\rho(t) = 0.06 \max(0, t-8)$ . Support validation and hysteresis reject harmful updates. Conformal intervals quantify context error, not rollout-failure probability.

#### V. EXPERIMENTS

Both methods use identical training data, K25 support, rank-8 adapters, optimization, and rollouts. The baseline is shared dynamics plus analytic projection and its matched adapter. Seed27 was untouched until the rule froze.

All 18 cells exceed 2%; the minimum confirmation is 2.0430%. Locked violation is zero and free-joint changes remain within 5%. Intact/D2/D4 object differences are at most 1.72%; maximum IID pusher difference is 1.081 mm. K curves are not monotonic, so we claim a frozen K25 rule only.

##### A. Real-arm protocol

[REAL-ROBOT TODO: Insert setup photo, calibration diagram, and results.] The frozen protocol uses a five-DoF ST3215 arm, eye-in-hand tracking, 10 Hz, intact/D2/D3, K=0/5/10/25, and three repetitions. Stops trigger on stale vision, feedback timeout, 50°C, raw current 400, or 3.5° lock drift.

#### VI. LIMITATIONS AND CONCLUSION

Current evidence is predictive, not closed-loop control. Diagnosis must provide lock identity and angle. Contact awareness precludes strict triangular causality. Tests cover single locks and planar pushing. Nevertheless, PC-RR-IPWM improves every held-out matched cell while preserving exact feasibility and controls. [REAL-ROBOT TODO: Add real-arm limitations and conclusion.]

TABLE III  
REAL-ARM PUSHING. [REAL-ROBOT TODO: POPULATE AFTER COLLECTION.]

| Lock   | Method      | Obj.H10 | Obj.H25 | Success |
|--------|-------------|---------|---------|---------|
| Intact | Shared/ours | —       | —       | —       |
| D2     | Shared/ours | —       | —       | —       |
| D3     | Shared/ours | —       | —       | —       |

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